Changes in Age and Racial/Ethnic Disparities in Anxiety and Depressive Symptoms During the COVID-19 Pandemic

Ronak Fathi

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1 1. Research Overview

1.1 1.1. Research question

This study examines how age-related and racial/ethnic disparities in symptoms consistent with anxiety or depression changed among U.S. adults during the COVID-19 pandemic and its aftermath.

The primary research question is:

How did differences in the estimated prevalence of anxiety or depressive symptoms across age groups and racial/ethnic groups evolve over the Household Pulse Survey period?

The analysis focuses on two demographic dimensions:

1. Age-related disparities.
2. Racial/ethnic disparities.

These are examined separately because the published aggregate dataset does not provide cross-classified age-by-race estimates.

1.2 1.2. Study aims

The primary aims are to:

- Describe national trends in the combined anxiety or depressive symptom indicator.
- Quantify differences between age groups over time.
- Quantify differences between racial/ethnic groups over time.
- Evaluate whether demographic gaps changed linearly or nonlinearly.
- Investigate documented survey transitions that may affect temporal comparability.
- Assess the robustness of findings under alternative model specifications and reference groups.

1.3 1.3. Outcomes

The primary outcome is the published percentage of adults reporting symptoms consistent with an anxiety disorder or depressive disorder.

The two separate indicators are treated as secondary outcomes:

- Symptoms of Anxiety Disorder.
- Symptoms of Depressive Disorder.

These indicators represent reported symptoms and should not be interpreted as clinical diagnoses.

2 2. Data Source and Study Design

2.1 2.1. Data source

The analysis uses published estimates from the U.S. Census Bureau Household Pulse Survey (HPS), disseminated through the Centers for Disease Control and Prevention and the National Center for Health Statistics.

The dataset contains estimates for national, state, and demographic subgroups across multiple survey periods.

The analysis uses published aggregate estimates rather than individual respondent-level records.

2.2 2.2. Study design

This is a repeated cross-sectional analysis of published survey estimates.

The same individuals are not followed over time. Changes in estimates therefore describe changes in population-level reported symptom prevalence, not within-person changes.

2.3 2.3. Data structure

The dataset includes:

- Indicator.
- Demographic group.
- State.

- Subgroup.
- Survey phase.
- Survey period.
- Survey-period start and end dates.
- Published estimate.
- Lower and upper confidence limits.

The survey periods are not equally spaced. Actual dates are therefore used in temporal analyses.

3 3. Statistical Analysis Plan

3.1 3.1. Primary estimands

For demographic subgroup g at survey time t, the primary estimand is the absolute prevalence difference relative to a reference group:

$$D_{g,t} = P_{g,t} - P_{\text{ref},t}$$

where:

- $P_{g,t}$ is the published prevalence estimate for subgroup g.
- $P_{\text{ref},t}$ is the published estimate for the reference subgroup.
- $D_{g,t}$ is the absolute difference in percentage points.

The relative measure is the prevalence ratio:

$$R_{g,t} = \frac{P_{g,t}}{P_{\text{ref},t}}$$

The log prevalence ratio is also calculated:

$$L_{g,t} = \log(R_{g,t})$$

Absolute differences are the primary disparity measure. Prevalence ratios and log prevalence ratios provide complementary descriptions.

3.2 3.2. Reference groups

For the primary analysis:

- Age reference: 60–69 years.
- Racial/ethnic reference: Non-Hispanic White, single race.

These categories are retained provisionally and examined through sensitivity analyses using alternative reference groups.

3.3 3.3. Candidate statistical models

The candidate models are fitted to the demographic gap trajectories, separately for age and race/ethnicity.

3.3.1 Model A: Linear trend

$$D_t = \beta_0 + \beta_1 t + \epsilon_t$$

This model estimates the average linear direction of change in the disparity.

3.3.2 Model B: Generalized additive model

$$D_t = \beta_0 + s(t) + \epsilon_t$$

The smooth function allows the disparity trajectory to change nonlinearly over time.

3.3.3 Model C: Interrupted time-series model

Where an HPS documentation-supported discontinuity is identified, the following segmented model is considered:

$$D_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t - t_0) I_t + \epsilon_t$$

Here, t_0 is the documented break date and I_t indicates observations after that date.

This model is not fitted unless a defensible break date has been established independently of the observed disparity results.

3.3.4 Model D: GAM with correlated errors

A GAM with a continuous-time autoregressive correlation structure is considered when residual diagnostics indicate temporal dependence.

This model retains a flexible nonlinear mean trajectory while accounting for correlation between observations.

3.4 3.4. Model comparison and selection

Candidate models are assessed using:

- Information criteria where likelihood comparisons are appropriate.
- Model complexity and effective degrees of freedom.
- Residual patterns.
- Residual autocorrelation.
- Sensitivity to influential observations.
- Interpretability and consistency with the research question.

The most complex model is not automatically selected.

The primary model is selected based on the combined evidence from model fit, diagnostics, temporal structure, and interpretability.

3.5 3.5. Uncertainty in published estimates

The dataset provides confidence intervals for individual subgroup estimates.

Approximate standard errors may be derived from confidence interval widths if the confidence level and construction are confirmed.

However, the covariance between subgroup estimates is not available in the current aggregate dataset. Consequently, the uncertainty of a difference between two subgroup estimates cannot be calculated exactly from the individual confidence intervals alone.

The primary model comparisons are therefore interpreted as analyses of published estimate trajectories. Model-based standard errors do not automatically represent survey-design-based uncertainty.

Any gap uncertainty calculated under an independence assumption is explicitly labelled as approximate and exploratory.

3.6 3.6. Multiple comparisons

The primary analysis includes multiple age-group and racial/ethnic-group comparisons.

Holm adjustment is used for families of related secondary inferential tests. The interpretation distinguishes between exploratory trajectory patterns and statistically supported evidence.

3.7 3.7. Sensitivity analyses

The planned sensitivity analyses include:

- Alternative analysis windows based on documented survey transitions.
- Alternative age reference groups.
- Alternative disparity measures.
- Linear versus nonlinear temporal specifications.
- Models with and without temporal correlation.
- Examination of unusually imprecise or influential subgroup estimates.

4 4. Data Preparation and Audit

4.1 4.1. Import and standardize the data

```
## Rows: 16,794
## Columns: 17
## $ indicator           <chr> "Symptoms of Depressive Disorder", "Symptoms of De~
## $ group               <chr> "National Estimate", "By Age", "By Age", "By Age",~
## $ state               <chr> "United States", "United States", "United States",~
## $ subgroup            <chr> "United States", "18 - 29 years", "30 - 39 years",~
## $ phase               <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", ~
## $ time_period         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ time_period_label   <chr> "Apr 23 - May 5, 2020", "Apr 23 - May 5, 2020", "A~
## $ start_date          <date> 2020-04-23, 2020-04-23, 2020-04-23, 2020-04-23, 2~
## $ end_date            <date> 2020-05-05, 2020-05-05, 2020-05-05, 2020-05-05, 2~
## $ value               <dbl> 23.5, 32.7, 25.7, 24.8, 23.2, 18.4, 13.6, 14.4, 20~
## $ low_ci              <dbl> 22.7, 30.2, 24.1, 23.3, 21.5, 17.0, 11.8, 9.0, 19.~
## $ high_ci             <dbl> 24.3, 35.2, 27.3, 26.2, 25.0, 19.7, 15.5, 21.4, 22~
## $ confidence_interval <chr> "22.7 - 24.3", "30.2 - 35.2", "24.1 - 27.3", "23.3~
## $ quartile_range      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ start_date_original <chr> "4/23/2020", "4/23/2020", "4/23/2020", "4/23/2020"~
## $ end_date_original   <chr> "5/5/2020", "5/5/2020", "5/5/2020", "5/5/2020", "5~
## $ time_numeric        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

4.2 4.2. Dataset overview

Table 1: Dataset overview

Measure	Value
Total records	16794
Unique indicators	3
Unique demographic groups	9
Unique subgroups	78
Unique survey periods	82
First survey date	2020-04-23
Last survey date	2024-09-16

4.3 4.3. Missing values

Table 2: Missing values by variable

Variable	Missing values
quartile_range	5777
value	707
low_ci	707
high_ci	707
confidence_interval	707

4.4 4.4. Duplicate records

```
## No duplicate indicator-group-subgroup-period records were detected.
```

4.5 4.5. Estimate and confidence interval checks

Table 3: Estimate and confidence interval validation

Minimum_estimate	Maximum_estimate	Estimates_below_zero	Estimates_above_100	Invalid_CI_order	Estimates_below_CI	Estimates_above_CI
4.6	85.2	0	0	0	0	0

4.6 4.6. Survey-period structure

Table 4: Spacing between survey periods

Number_of_periods	Minimum_gap_days	Median_gap_days	Mean_gap_days	Maximum_gap_days
82	6	14	19.5	50

4.7 4.7. Phase distribution

Table 5: Survey phase distribution for the primary outcome

phase	Number of periods
-1	10
1	12
2	5
3.0 (Jan 6 - Mar 29)	6
3.0 (Oct 28 - Dec 21)	4
3.1	9
3.2	6
3.3	3
3.4	3
3.5	3
3.6	3
3.7	3
3.8	3
3.9	3
4	3

phase	Number of periods
4.1	4
4.2	2

4.8 4.8. Period completeness across indicators

The primary outcome is used as the reference calendar for each demographic subgroup. Secondary indicators are checked against the same subgroup-specific periods.

This distinguishes between periods with no record and periods with a record but a missing estimate.

Table 6: Completeness of secondary indicators relative to the primary outcome

indicator	group	status	Number of periods
Symptoms of Anxiety Disorder	By Age	Estimate available	504
Symptoms of Anxiety Disorder	By Age	Record but missing estimate	70
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	Estimate available	504
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	Record but missing estimate	70
Symptoms of Depressive Disorder	By Age	Estimate available	504
Symptoms of Depressive Disorder	By Age	Record but missing estimate	70
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Estimate available	360
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Record but missing estimate	50
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Estimate available	360
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Record but missing estimate	50
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Estimate available	360
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Record but missing estimate	50

Table 7: Period completeness by demographic subgroup

indicator	group	subgroup	Expected_periods	Available_estimates	Missing_records	Missing_estimates
Symptoms of Anxiety Disorder	By Age	18 - 29 years	82	72	0	10
Symptoms of Anxiety Disorder	By Age	30 - 39 years	82	72	0	10
Symptoms of Anxiety Disorder	By Age	40 - 49 years	82	72	0	10
Symptoms of Anxiety Disorder	By Age	50 - 59 years	82	72	0	10

indicator	group	subgroup	Expected_periods	Available_estimates	Missing_reasons	Missing_estimates
Symptoms of Anxiety Disorder	By Age	60 - 69 years	82	72	0	10
Symptoms of Anxiety Disorder	By Age	70 - 79 years	82	72	0	10
Symptoms of Anxiety Disorder	By Age	80 years and above	82	72	0	10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Hispanic or Latino	82	72	0	10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	82	72	0	10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic Black, single race	82	72	0	10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic White, single race	82	72	0	10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	18 - 29 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	30 - 39 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	40 - 49 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	50 - 59 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	60 - 69 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	70 - 79 years	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	80 years and above	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Hispanic or Latino	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Black, single race	82	72	0	10

indicator	group	subgroup	Expected_periods	Available_estimates	Missing_records	Missing_estimates
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic White, single race	82	72	0	10
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	82	72	0	10
Symptoms of Depressive Disorder	By Age	18 - 29 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	30 - 39 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	40 - 49 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	50 - 59 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	60 - 69 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	70 - 79 years	82	72	0	10
Symptoms of Depressive Disorder	By Age	80 years and above	82	72	0	10
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Hispanic or Latino	82	72	0	10
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	82	72	0	10
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Black, single race	82	72	0	10
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic White, single race	82	72	0	10
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	82	72	0	10

4.9 4.9. Identify missing secondary-outcome periods

Table 8: Examples of missing secondary-outcome periods

indicator	group	subgroup	start_date	end_date	status
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2020-07-22	2020-08-18	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2020-12-22	2021-01-05	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2021-03-30	2021-04-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2021-07-06	2021-07-20	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2021-10-12	2021-11-30	Record but missing estimate

indicator	group	subgroup	start_date	end_date	status
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2022-02-23	2022-03-01	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2022-05-25	2022-05-31	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2022-08-24	2022-09-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2022-11-30	2022-12-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	18 - 29 years	2023-11-27	2024-01-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2020-07-22	2020-08-18	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2020-12-22	2021-01-05	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2021-03-30	2021-04-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2021-07-06	2021-07-20	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2021-10-12	2021-11-30	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2022-02-23	2022-03-01	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2022-05-25	2022-05-31	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2022-08-24	2022-09-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2022-11-30	2022-12-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	30 - 39 years	2023-11-27	2024-01-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2020-07-22	2020-08-18	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2020-12-22	2021-01-05	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2021-03-30	2021-04-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2021-07-06	2021-07-20	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2021-10-12	2021-11-30	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2022-02-23	2022-03-01	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2022-05-25	2022-05-31	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2022-08-24	2022-09-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2022-11-30	2022-12-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	40 - 49 years	2023-11-27	2024-01-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2020-07-22	2020-08-18	Record but missing estimate

indicator	group	subgroup	start_date	end_date	status
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2020-12-22	2021-01-05	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2021-03-30	2021-04-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2021-07-06	2021-07-20	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2021-10-12	2021-11-30	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2022-02-23	2022-03-01	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2022-05-25	2022-05-31	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2022-08-24	2022-09-13	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2022-11-30	2022-12-08	Record but missing estimate
Symptoms of Anxiety Disorder	By Age	50 - 59 years	2023-11-27	2024-01-08	Record but missing estimate

4.10 4.10. Summary by demographic group and indicator

Table 9: Summary of demographic estimates by indicator

indicator	group	Records	Subgroups	Periods	Available_estimates	Mean_estimate
Symptoms of Anxiety Disorder	By Age	574	7	82	504	26.10
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	410	5	82	360	29.18
Symptoms of Anxiety Disorder or Depressive Disorder	By Age	574	7	82	504	30.52
Symptoms of Anxiety Disorder or Depressive Disorder	By Race/Hispanic ethnicity	410	5	82	360	34.26
Symptoms of Depressive Disorder	By Age	574	7	82	504	21.06
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	410	5	82	360	24.16

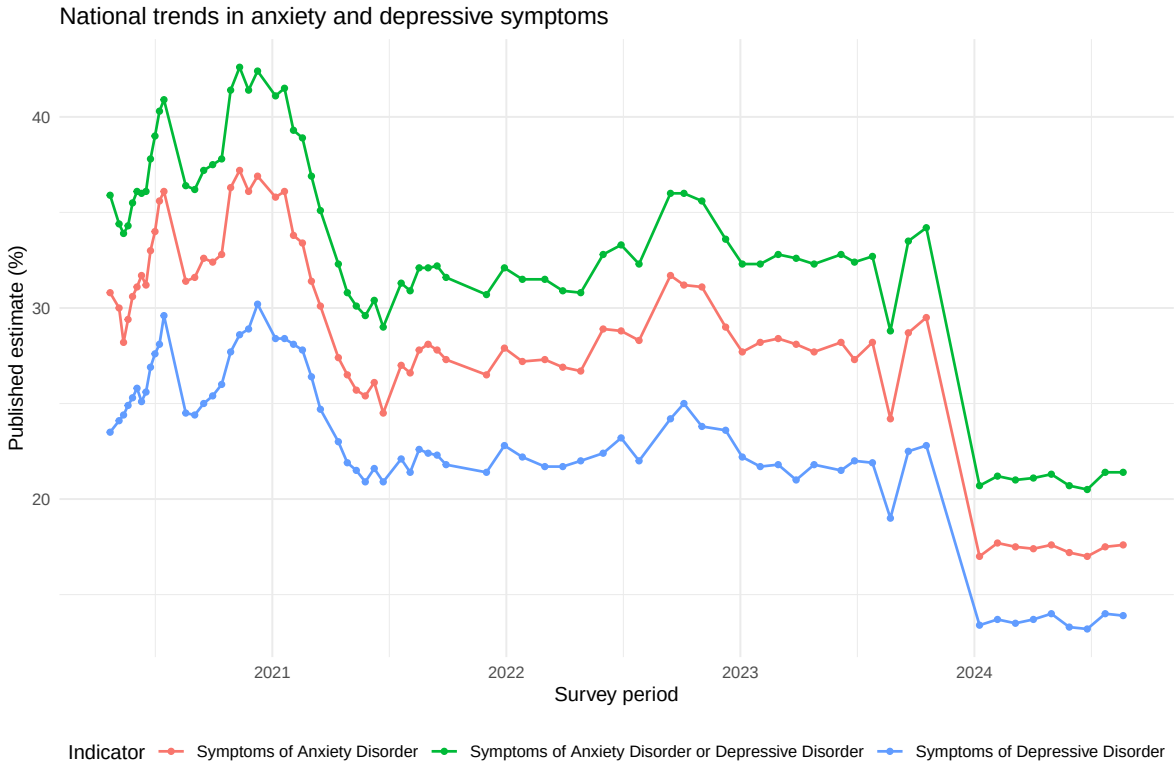
5 5. National Trends

5.1 5.1. National estimates

Table 10: National prevalence estimates

indicator	Number_of_periods	Mean	SD	Minimum	Maximum
Symptoms of Anxiety Disorder	72	28.39	5.21	17.0	37.2
Symptoms of Anxiety Disorder or Depressive Disorder	72	32.94	5.66	20.5	42.6
Symptoms of Depressive Disorder	72	22.70	4.23	13.2	30.2

5.2 5.2. National trend visualization



6 6. Construction of Demographic Disparity Measures

6.1 6.1. Prepare age estimates

Table 11: Age disparities relative to the reference group

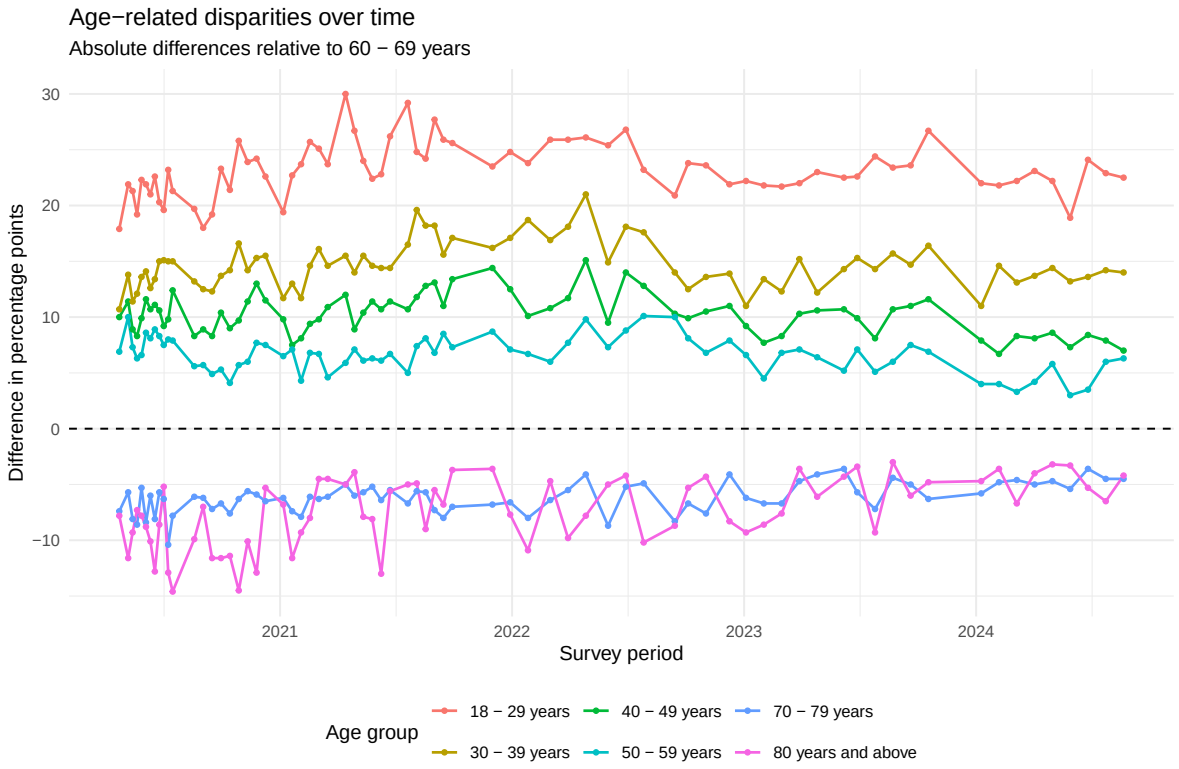
subgroup	Mean_gap	Mean_ratio	Number_of_periods
18 - 29 years	23.19	2.01	72
30 - 39 years	14.64	1.63	72
40 - 49 years	10.26	1.44	72
50 - 59 years	6.65	1.28	72
70 - 79 years	-6.20	0.74	72
80 years and above	-7.39	0.70	72

6.2 6.2. Prepare racial/ethnic estimates

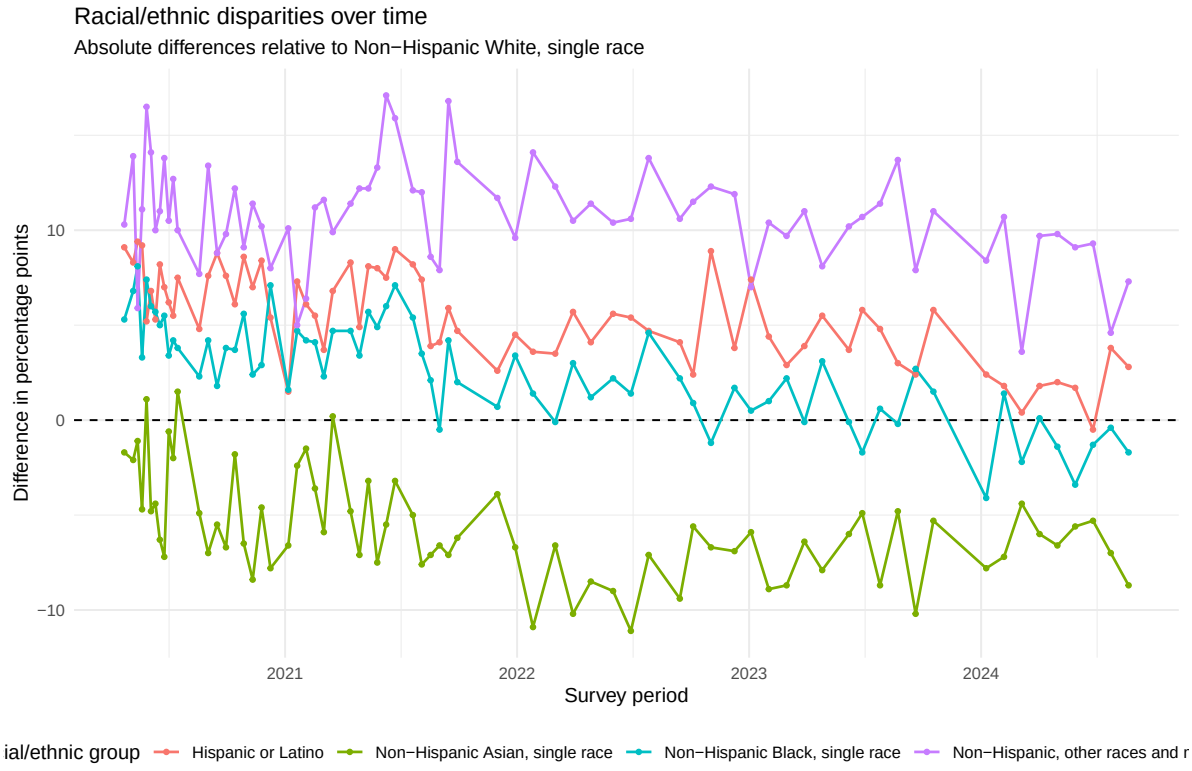
Table 12: Racial/ethnic disparities relative to the reference group

subgroup	Mean_gap	Mean_ratio	Number_of_periods
Hispanic or Latino	5.38	1.17	72
Non-Hispanic Asian, single race	-5.77	0.81	72
Non-Hispanic Black, single race	2.59	1.08	72
Non-Hispanic, other races and multiple races	10.72	1.35	72

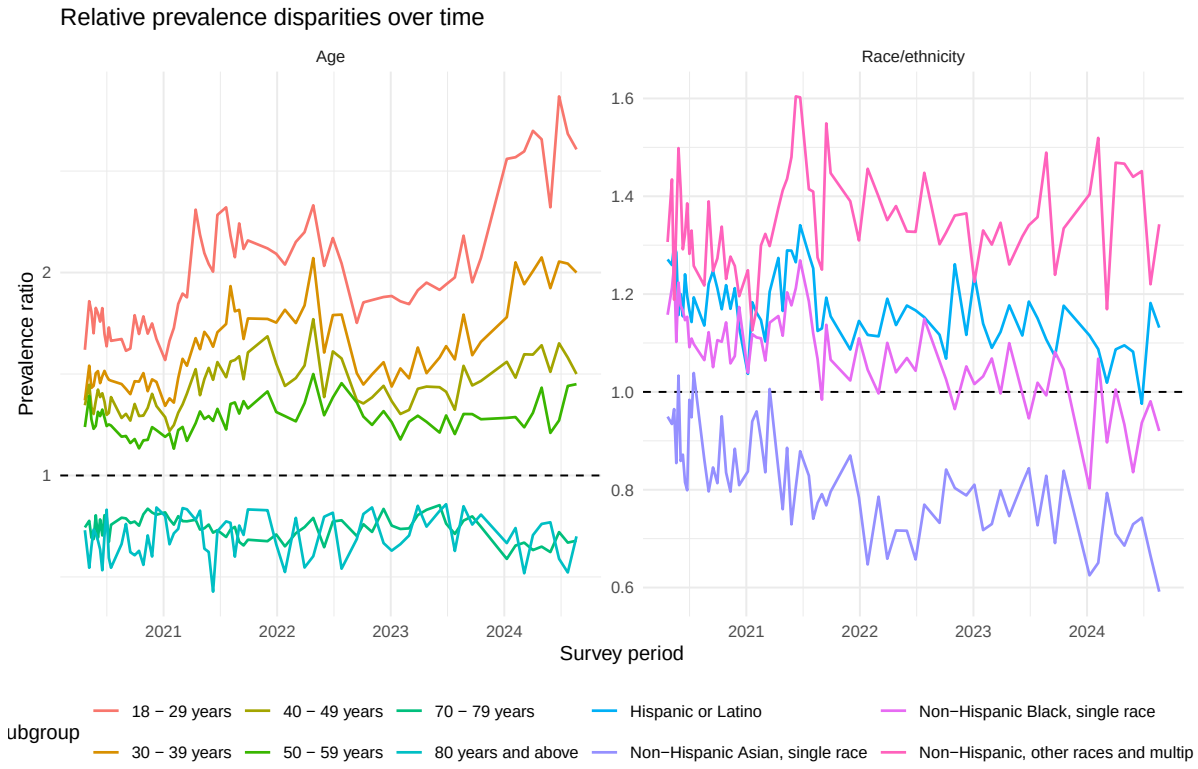
6.3 6.3. Plot raw age disparity trajectories



6.4 6.4. Plot raw racial/ethnic disparity trajectories



6.5 6.5. Relative disparity trajectories



7 7. Primary Statistical Modelling

7.1 7.1. Modelling strategy

The candidate models are fitted separately to each demographic gap trajectory.

The response variable is the absolute prevalence difference in percentage points.

The time variable is measured in days from the first observation in the relevant series. This preserves the irregular spacing of the survey periods.

The candidate models are:

- Linear trend model.
- Generalized additive model.
- Interrupted time-series model, only when a documented break date is available.
- GAM with continuous-time correlated errors.

The model comparison is based on the same observations within each subgroup series.

7.2 7.2. Prepare model datasets

Table 13: Observations available for each disparity model

dimension	subgroup	Number_of_observations	First_date	Last_date	Mean_gap	SD_gap
Age	18 - 29 years	72	2020-04-23	2024-08-20	23.19	2.41

dimension	subgroup	Number_of_observations	First_date	Last_date	Mean_gap	SD_gap
Age	30 - 39 years	72	2020-04-23	2024-08-20	14.64	2.07
Age	40 - 49 years	72	2020-04-23	2024-08-20	10.26	1.81
Age	50 - 59 years	72	2020-04-23	2024-08-20	6.65	1.60
Age	70 - 79 years	72	2020-04-23	2024-08-20	-6.20	1.34
Age	80 years and above	72	2020-04-23	2024-08-20	-7.39	3.06
Race/ethnicity	Hispanic or Latino	72	2020-04-23	2024-08-20	5.38	2.37
Race/ethnicity	Non-Hispanic Asian, single race	72	2020-04-23	2024-08-20	-5.77	2.70
Race/ethnicity	Non-Hispanic Black, single race	72	2020-04-23	2024-08-20	2.59	2.67
Race/ethnicity	Non-Hispanic, other races and multiple races	72	2020-04-23	2024-08-20	10.72	2.66

7.3 7.3. Fit candidate models

The following function fits the linear, GAM, and correlated-error GAM models to one disparity trajectory.

The correlated-error model uses a continuous-time AR(1) structure because the survey dates are irregularly spaced.

Table 14: Candidate model fitting status

series	linear_fitted	gam_fitted	gamm_fitted
Age | 18 - 29 years	TRUE	TRUE	TRUE
Age | 30 - 39 years	TRUE	TRUE	TRUE
Age | 40 - 49 years	TRUE	TRUE	TRUE
Age | 50 - 59 years	TRUE	TRUE	TRUE
Age | 70 - 79 years	TRUE	TRUE	TRUE
Age | 80 years and above	TRUE	TRUE	TRUE
Race/ethnicity | Hispanic or Latino	TRUE	TRUE	TRUE
Race/ethnicity | Non-Hispanic Asian, single race	TRUE	TRUE	TRUE
Race/ethnicity | Non-Hispanic Black, single race	TRUE	TRUE	TRUE
Race/ethnicity | Non-Hispanic, other races and multiple races	TRUE	TRUE	TRUE

7.4 7.4. Linear trend results

Table 15: Linear trend estimates per 30 days

series	slope_per_30_days	lower_30_days	upper_30_days	p.value
Age | 18 - 29 years	0.0151	-0.0202	0.0505	0.3961
Age | 30 - 39 years	0.0024	-0.0281	0.0329	0.8751
Age | 40 - 49 years	-0.0316	-0.0572	-0.0060	0.0162
Age | 50 - 59 years	-0.0330	-0.0551	-0.0108	0.0041
Age | 70 - 79 years	0.0435	0.0267	0.0603	0.0000

series	slope_per_30_days	lower_30_days	upper_30_days	p.value
Age | 80 years and above	0.1036	0.0659	0.1412	0.0000
Race/ethnicity | Hispanic or Latino	-0.1015	-0.1266	-0.0764	0.0000
Race/ethnicity | Non-Hispanic Asian, single race	-0.0760	-0.1112	-0.0407	0.0001
Race/ethnicity | Non-Hispanic Black, single race	-0.1290	-0.1533	-0.1046	0.0000
Race/ethnicity | Non-Hispanic, other races and multiple races	-0.0472	-0.0847	-0.0098	0.0142

7.5 7.5. GAM results

Table 16: GAM smooth-term results with Holm adjustment

series	effective_degrees_freedom	reference_degrees_freedom	test_statistic	p_value	adjusted_p_value
Age | 18 - 29 years	4.2272	5.0123	10.5950	0.000	0.000
Age | 30 - 39 years	5.6982	6.3346	10.9497	0.000	0.000
Age | 40 - 49 years	4.2757	5.0618	9.2596	0.000	0.000
Age | 50 - 59 years	4.7559	5.5315	9.0664	0.000	0.000
Age | 70 - 79 years	1.0004	1.0008	26.6241	0.000	0.000
Age | 80 years and above	1.3396	1.6017	20.3432	0.000	0.000
Race/ethnicity | Hispanic or Latino	1.0001	1.0003	65.0317	0.000	0.000
Race/ethnicity | Non-Hispanic Asian, single race	2.8322	3.4658	10.4561	0.000	0.000
Race/ethnicity | Non-Hispanic Black, single race	1.0002	1.0003	111.7495	0.000	0.000
Race/ethnicity | Non-Hispanic, other races and multiple races	2.3877	2.9431	4.1398	0.009	0.009

7.6 7.6. Interrupted time-series analysis

The interrupted time-series model is fitted only when a break date has been documented from the survey methodology.

Do not enter a date based only on a visual change in the outcome.

Table 17: Interrupted time-series model fitting status

series	fitted
Age | 18 - 29 years	FALSE
Age | 30 - 39 years	FALSE
Age | 40 - 49 years	FALSE
Age | 50 - 59 years	FALSE
Age | 70 - 79 years	FALSE
Age | 80 years and above	FALSE
Race/ethnicity | Hispanic or Latino	FALSE
Race/ethnicity | Non-Hispanic Asian, single race	FALSE
Race/ethnicity | Non-Hispanic Black, single race	FALSE
Race/ethnicity | Non-Hispanic, other races and multiple races	FALSE

```
## No interrupted time-series models were fitted.  A documented survey discontinuity date has not been s
```

7.7 7.7. Model comparison

AICc is calculated from the fitted model likelihoods. Comparisons are made only within the same subgroup series and using the same observations.

The GAMM likelihood is taken from its underlying mixed-model component.

Table 18: Candidate model comparison using AICc

series	model	AICc	delta_AICc	AICc_rank
Age | 18 - 29 years	Linear	335.89	26.71	2
Age | 18 - 29 years	GAMM_CAR1	309.18	0.00	1
Age | 30 - 39 years	Linear	314.70	36.54	2
Age | 30 - 39 years	GAMM_CAR1	278.15	0.00	1
Age | 40 - 49 years	Linear	289.41	16.75	2
Age | 40 - 49 years	GAMM_CAR1	272.67	0.00	1
Age | 50 - 59 years	Linear	268.59	19.10	2
Age | 50 - 59 years	GAMM_CAR1	249.49	0.00	1
Age | 70 - 79 years	Linear	229.02	0.00	1
Age | 70 - 79 years	GAMM_CAR1	235.66	6.64	2
Age | 80 years and above	Linear	345.02	0.00	1
Age | 80 years and above	GAMM_CAR1	348.40	3.38	2
Race/ethnicity | Hispanic or Latino	Linear	286.67	0.00	1
Race/ethnicity | Hispanic or Latino	GAMM_CAR1	290.97	4.30	2
Race/ethnicity | Non-Hispanic Asian, single race	Linear	335.63	5.86	2
Race/ethnicity | Non-Hispanic Asian, single race	GAMM_CAR1	329.77	0.00	1
Race/ethnicity | Non-Hispanic Black, single race	Linear	282.18	0.00	1
Race/ethnicity | Non-Hispanic Black, single race	GAMM_CAR1	287.34	5.17	2
Race/ethnicity | Non-Hispanic, other races and multiple races	Linear	344.37	0.00	1
Race/ethnicity | Non-Hispanic, other races and multiple races	GAMM_CAR1	344.51	0.14	2

7.8 7.8. Model selection summary

The model with the lowest AICc is identified as the best-fitting candidate among the models successfully fitted to the same series.

This is a diagnostic aid, not an automatic selection rule. Residual behaviour, temporal dependence, model complexity, and interpretability must also be considered.

Table 19: Best-fitting candidate according to AICc

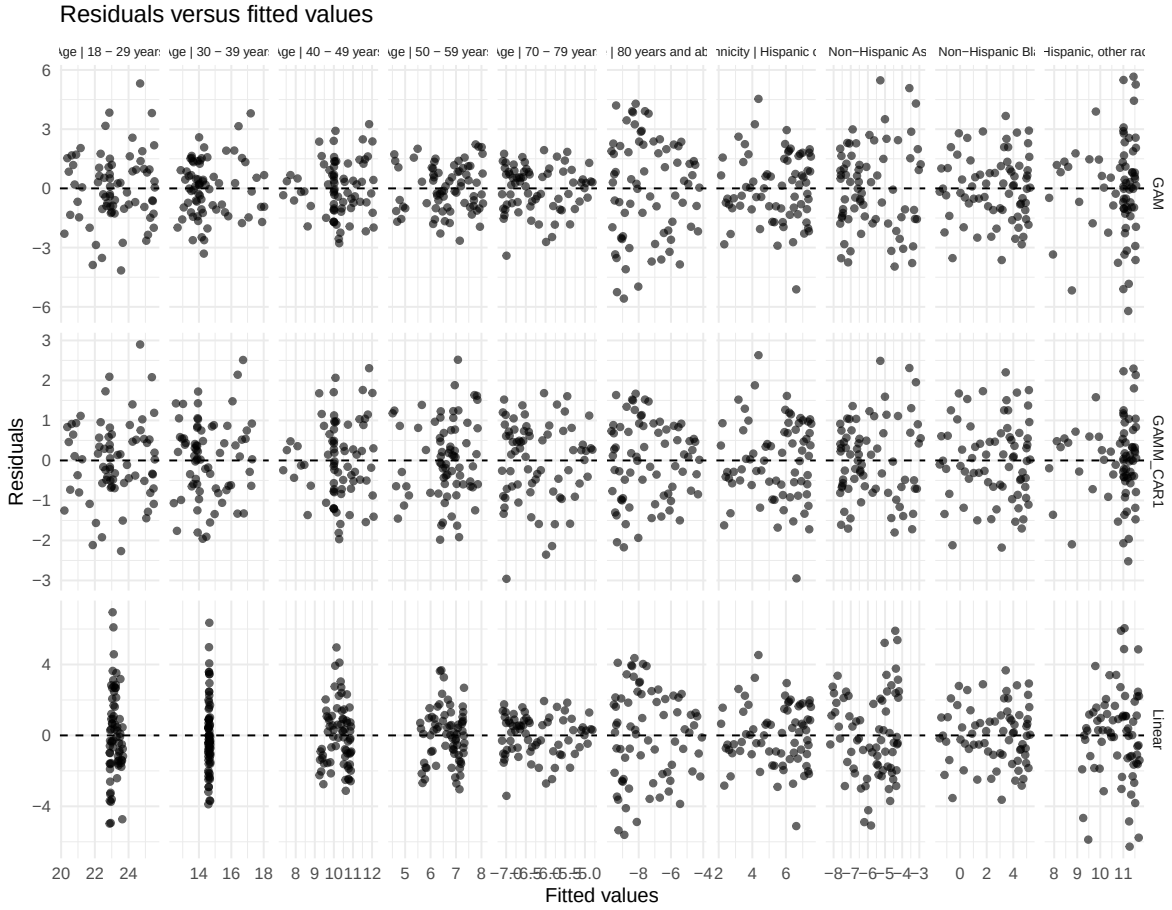
series	Selected_by_AICc	AICc	delta_AICc
Age | 18 - 29 years	GAMM_CAR1	309.18	0
Age | 30 - 39 years	GAMM_CAR1	278.15	0
Age | 40 - 49 years	GAMM_CAR1	272.67	0
Age | 50 - 59 years	GAMM_CAR1	249.49	0
Age | 70 - 79 years	Linear	229.02	0
Age | 80 years and above	Linear	345.02	0
Race/ethnicity | Hispanic or Latino	Linear	286.67	0
Race/ethnicity | Non-Hispanic Asian, single race	GAMM_CAR1	329.77	0
Race/ethnicity | Non-Hispanic Black, single race	Linear	282.18	0
Race/ethnicity | Non-Hispanic, other races and multiple races	Linear	344.37	0

8 8. Model Diagnostics

8.1 8.1. Residual diagnostics

Residual diagnostics are examined for the linear, GAM, and correlated-error GAM models.

8.2 8.2. Residual plots

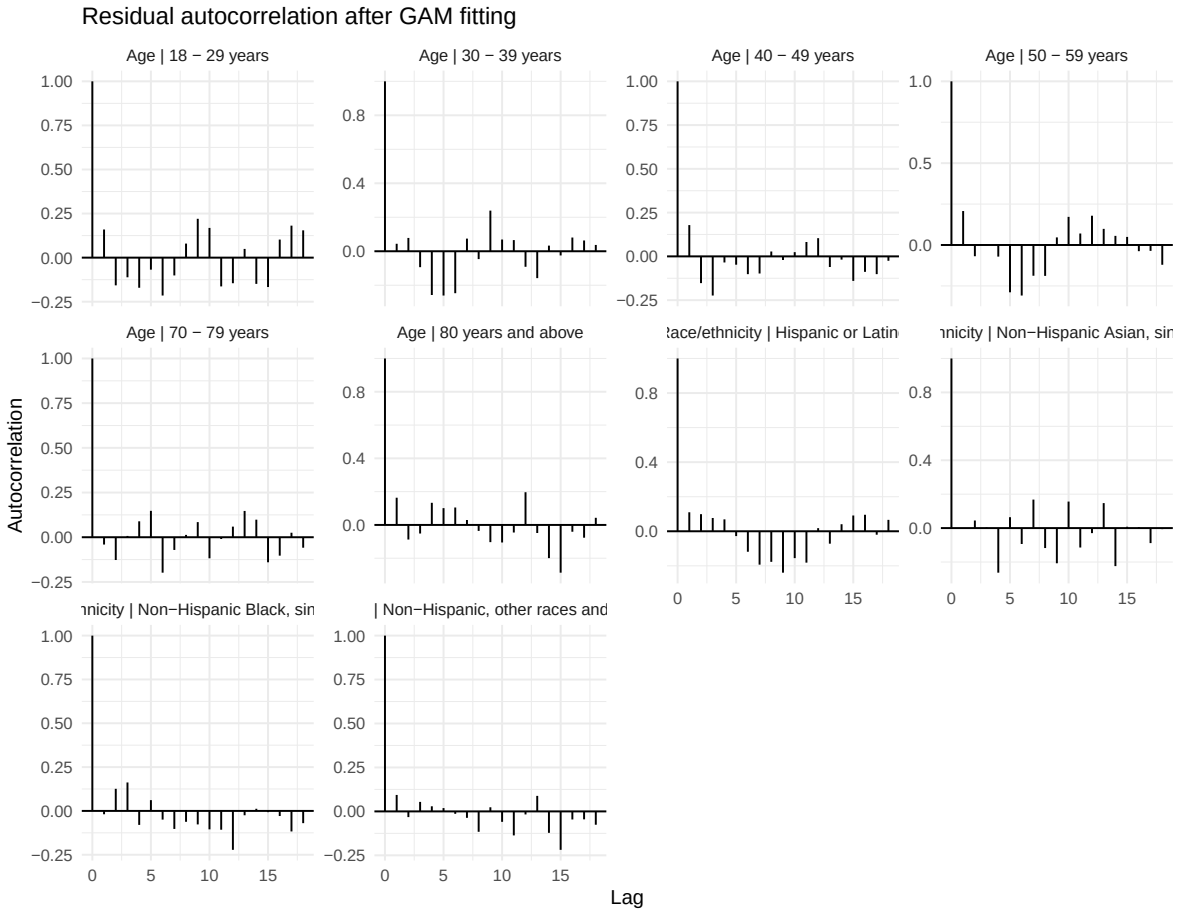


8.3 8.3. Residual autocorrelation

Table 20: Residual summary by candidate model

series	model	Residual_SD	Mean_residual
Age | 18 - 29 years	GAM	1.778	0.000
Age | 18 - 29 years	GAMM_CAR1	0.970	0.000
Age | 18 - 29 years	Linear	2.402	0.000
Age | 30 - 39 years	GAM	1.424	0.000
Age | 30 - 39 years	GAMM_CAR1	0.968	0.035
Age | 30 - 39 years	Linear	2.074	0.000
Age | 40 - 49 years	GAM	1.368	0.000
Age | 40 - 49 years	GAMM_CAR1	0.969	0.000
Age | 40 - 49 years	Linear	1.740	0.000
Age | 50 - 59 years	GAM	1.186	0.000
Age | 50 - 59 years	GAMM_CAR1	0.973	0.016

series	model	Residual_SD	Mean_residual
Age | 50 - 59 years	Linear	1.506	0.000
Age | 70 - 79 years	GAM	1.144	0.000
Age | 70 - 79 years	GAMM_CAR1	0.993	0.000
Age | 70 - 79 years	Linear	1.144	0.000
Age | 80 years and above	GAM	2.546	0.000
Age | 80 years and above	GAMM_CAR1	0.991	0.000
Age | 80 years and above	Linear	2.560	0.000
Race/ethnicity | Hispanic or Latino	GAM	1.707	0.000
Race/ethnicity | Hispanic or Latino	GAMM_CAR1	0.993	0.000
Race/ethnicity | Hispanic or Latino	Linear	1.707	0.000
Race/ethnicity | Non-Hispanic Asian, single race	GAM	2.155	0.000
Race/ethnicity | Non-Hispanic Asian, single race	GAMM_CAR1	0.980	0.000
Race/ethnicity | Non-Hispanic Asian, single race	Linear	2.398	0.000
Race/ethnicity | Non-Hispanic Black, single race	GAM	1.655	0.000
Race/ethnicity | Non-Hispanic Black, single race	GAMM_CAR1	0.993	0.000
Race/ethnicity | Non-Hispanic Black, single race	Linear	1.655	0.000
Race/ethnicity | Non-Hispanic, other races and multiple races	GAM	2.422	0.000
Race/ethnicity | Non-Hispanic, other races and multiple races	GAMM_CAR1	0.983	0.000
Race/ethnicity | Non-Hispanic, other races and multiple races	Linear	2.548	0.000

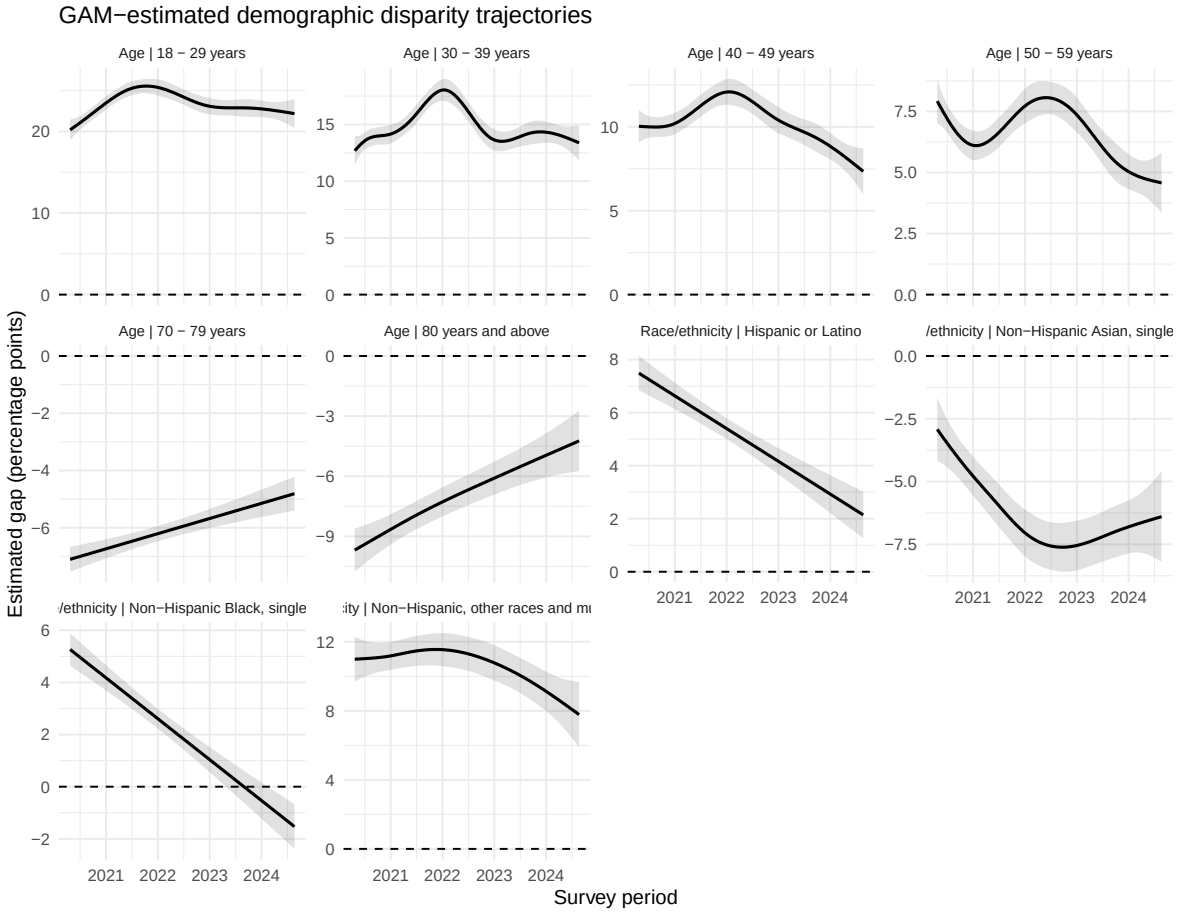


9 9. Estimated Disparity Trajectories

9.1 9.1. GAM predictions

The following plots display the estimated smooth trajectories from the GAM candidate models.

The intervals are model-based confidence intervals for the fitted mean trajectory. They are not survey-design-based confidence intervals for the underlying population disparity.



10 10. Secondary Outcome Analyses

The separate anxiety and depression indicators are analysed as secondary outcomes.

The primary modelling framework is not automatically repeated for every secondary outcome. Their main purpose is to assess whether the patterns observed for the combined indicator are also evident in the separate symptom indicators.

10.1 10.1. Prepare secondary outcome estimates

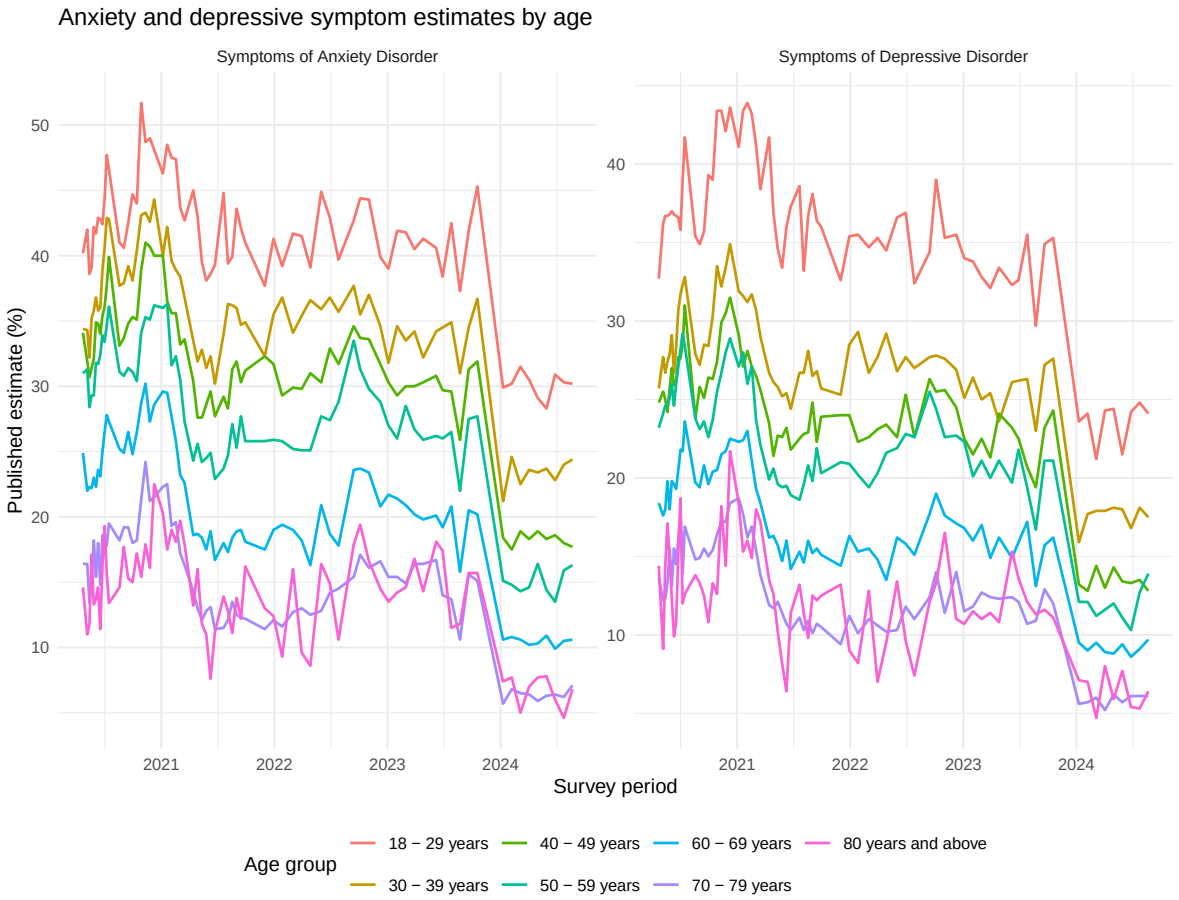
Table 21: Descriptive summary of secondary outcomes

indicator	group	subgroup	Number_of_periods	Mean_estimate	SD_estimate	Minimum	Maximum
Symptoms of Anxiety Disorder	By Age	18 - 29 years	72	41.00	5.11	28.3	51.7

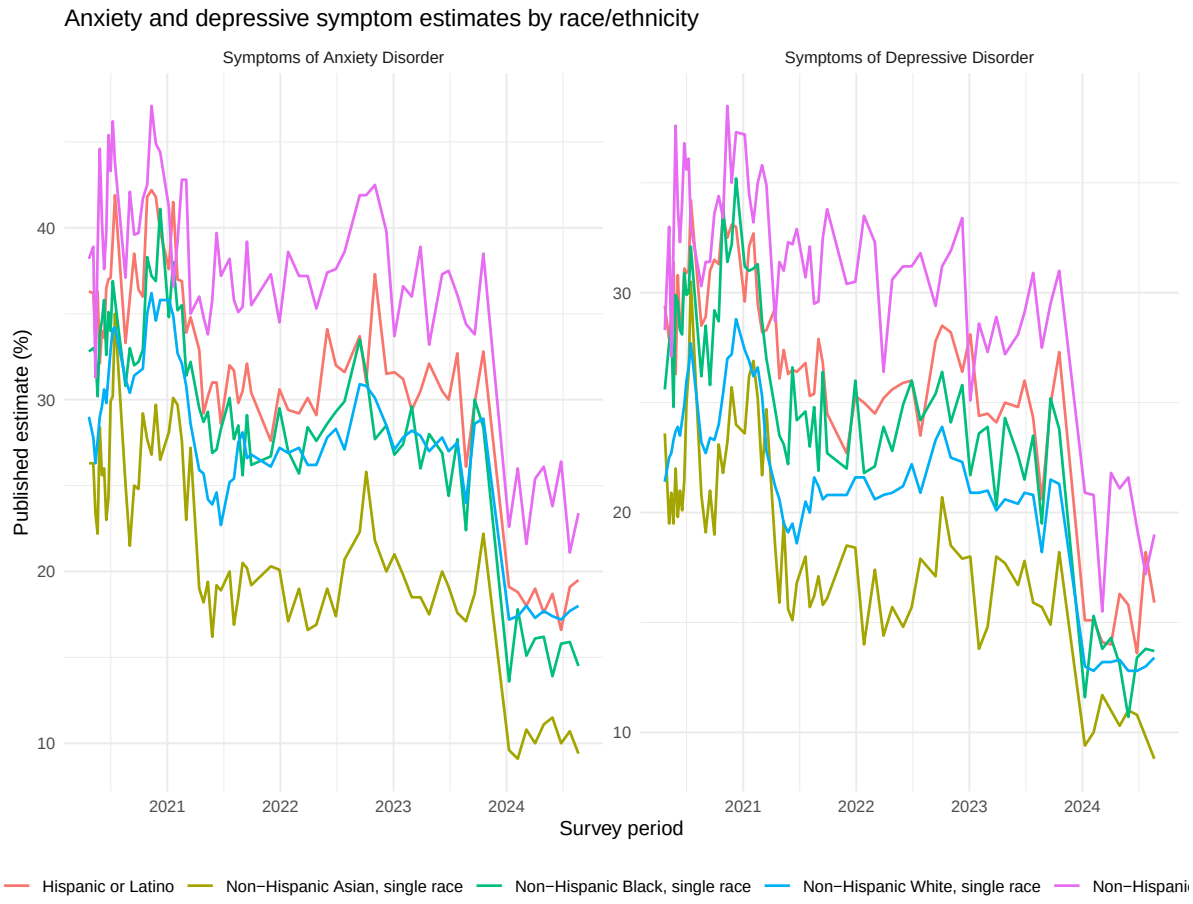
indicator	group	subgroup	Number_of_periods	Mean_estimate	SD_estimate	Minimum	Maximum
Symptoms of Anxiety Disorder	By Age	30 - 39 years	72	34.70	5.34	21.2	44.3
Symptoms of Anxiety Disorder	By Age	40 - 49 years	72	30.83	5.78	17.5	41.0
Symptoms of Anxiety Disorder	By Age	50 - 59 years	72	27.21	5.83	13.5	36.3
Symptoms of Anxiety Disorder	By Age	60 - 69 years	72	20.60	5.23	9.9	30.2
Symptoms of Anxiety Disorder	By Age	70 - 79 years	72	14.62	4.34	5.7	24.2
Symptoms of Anxiety Disorder	By Age	80 years and above	72	13.75	3.95	4.6	22.5
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Hispanic or Latino	72	31.68	6.21	16.6	42.2
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	72	20.94	5.83	9.1	35.0
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic Black, single race	72	28.84	6.28	13.6	41.1
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic White, single race	72	27.57	4.87	17.2	36.2
Symptoms of Anxiety Disorder	By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	72	36.85	5.93	21.1	47.1
Symptoms of Depressive Disorder	By Age	18 - 29 years	72	35.09	5.40	21.2	43.9
Symptoms of Depressive Disorder	By Age	30 - 39 years	72	26.69	4.24	15.9	34.9
Symptoms of Depressive Disorder	By Age	40 - 49 years	72	23.35	4.49	12.8	31.5
Symptoms of Depressive Disorder	By Age	50 - 59 years	72	21.49	4.62	10.3	29.2
Symptoms of Depressive Disorder	By Age	60 - 69 years	72	16.67	3.81	8.6	23.6
Symptoms of Depressive Disorder	By Age	70 - 79 years	72	12.25	3.36	5.2	18.7
Symptoms of Depressive Disorder	By Age	80 years and above	72	11.90	3.53	4.7	21.7
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Hispanic or Latino	72	26.23	5.02	13.6	34.2

indicator	group	subgroup	Number_of_periods	Mean_estimate	SD_estimate	Minimum	Maximum
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	72	18.22	4.63	8.8	30.5
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic Black, single race	72	24.58	5.46	10.7	35.2
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic White, single race	72	21.40	3.94	12.8	28.8
Symptoms of Depressive Disorder	By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	72	30.37	4.96	15.5	38.5

10.2 10.2. Secondary outcome trends by age



10.3 10.3. Secondary outcome trends by race/ethnicity



11 11. Sensitivity Analyses

11.1 11.1. Alternative age reference group

The 50–59 age group is used as an alternative reference category.

Table 22: Age disparities using 50–59 years as the reference

subgroup	Mean_gap	SD_gap	Number_of_periods
18 - 29 years	16.54	2.75	72
30 - 39 years	7.99	2.12	72
40 - 49 years	3.61	1.46	72
60 - 69 years	-6.65	1.60	72
70 - 79 years	-12.85	2.29	72
80 years and above	-14.04	3.74	72

11.2 11.2. Alternative disparity measures

The absolute difference, prevalence ratio, and log prevalence ratio are compared descriptively.

Table 23: Comparison of alternative disparity measures

dimension	subgroup	Mean_absolute_gap	Mean_prevalence_ratio	Median_log_prevalence_ratio
Age	18 - 29 years	23.189	2.006	0.685
Age	30 - 39 years	14.636	1.635	0.484
Age	40 - 49 years	10.256	1.437	0.360
Age	50 - 59 years	6.646	1.280	0.245
Age	70 - 79 years	-6.200	0.740	-0.304
Age	80 years and above	-7.392	0.699	-0.370
Race/ethnicity	Hispanic or Latino	5.383	1.168	0.153
Race/ethnicity	Non-Hispanic Asian, single race	-5.765	0.809	-0.219
Race/ethnicity	Non-Hispanic Black, single race	2.587	1.075	0.069
Race/ethnicity	Non-Hispanic, other races and multiple races	10.722	1.347	0.295

11.3 11.3. Alternative analysis windows

The following function allows the analysis to be repeated within a specified date window.

The window should be chosen using HPS documentation, not based on which dates produce statistically significant results.

The primary analysis uses the full available period. Once survey documentation has been reviewed, documented transitions can be entered below.

```
## No alternative analysis window has been specified.  The primary analysis retains the full available p
```

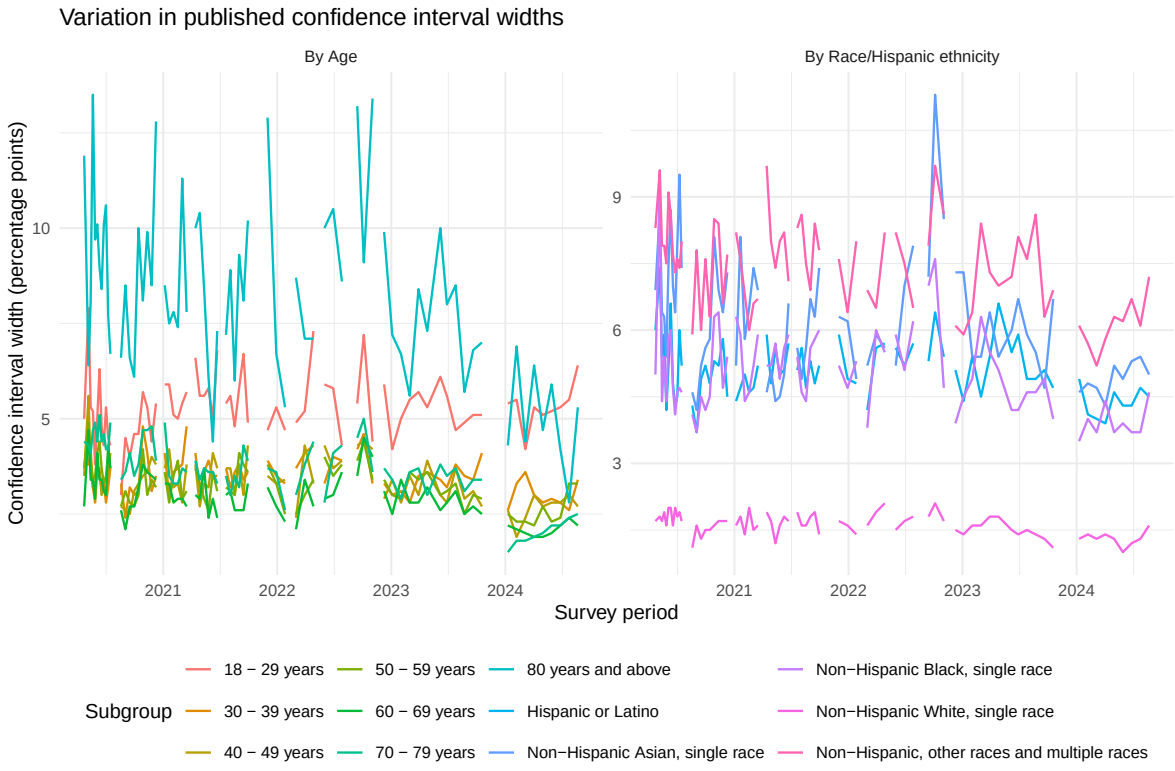
11.4 11.4. Investigate imprecise subgroup estimates

Subgroups with unusually wide confidence intervals are not automatically excluded.

Their precision is examined before deciding whether any exclusion is methodologically justified.

Table 24: Published confidence interval widths by demographic subgroup

group	subgroup	Number_of_periods	Mean_CI_width	Median_CI_width	Maximum_CI_width
By Age	80 years and above	72	8.12	8.05	13.5
By Race/Hispanic ethnicity	Non-Hispanic, other races and multiple races	72	7.40	7.50	9.7
By Race/Hispanic ethnicity	Non-Hispanic Asian, single race	72	6.21	5.95	11.3
By Age	18 - 29 years	72	5.29	5.30	7.9
By Race/Hispanic ethnicity	Hispanic or Latino	72	5.10	5.10	7.4
By Race/Hispanic ethnicity	Non-Hispanic Black, single race	72	5.03	4.80	9.3
By Age	70 - 79 years	72	3.62	3.60	5.1
By Age	30 - 39 years	72	3.49	3.50	4.8
By Age	40 - 49 years	72	3.37	3.30	5.6
By Age	50 - 59 years	72	3.29	3.30	4.9
By Age	60 - 69 years	72	2.96	2.90	4.7
By Race/Hispanic ethnicity	Non-Hispanic White, single race	72	1.61	1.60	2.1



11.5 11.5. Approximate uncertainty of subgroup differences

If the published confidence intervals are confirmed to be 95% intervals constructed using an approximately normal method, individual standard errors can be approximated.

The following calculation assumes independence between the subgroup estimates. This assumption may not hold for estimates derived from the same survey.

Therefore, these calculations are exploratory and must not be presented as exact survey-design-based confidence intervals for disparities.

Table 25: Exploratory uncertainty estimates under an independence assumption

dimension	subgroup	Number_of_periods	Mean_approximate_SE	Proportion_intervals_excluding_zero
Age	18 - 29 years	72	1.554	1.000
Age	30 - 39 years	72	1.173	1.000
Age	40 - 49 years	72	1.146	1.000
Age	50 - 59 years	72	1.131	1.000
Age	70 - 79 years	72	1.197	1.000
Age	80 years and above	72	2.212	0.792
Race/ethnicity	Hispanic or Latino	72	1.365	0.847
Race/ethnicity	Non-Hispanic Asian, single race	72	1.638	0.806
Race/ethnicity	Non-Hispanic Black, single race	72	1.349	0.542
Race/ethnicity	Non-Hispanic, other races and multiple races	72	1.932	1.000

12 12. Interpretation and Limitations

12.1 12.1. Interpretation of demographic disparities

The absolute difference describes the difference in published prevalence estimates in percentage points.

A positive gap indicates that the subgroup estimate is higher than the reference-group estimate. A negative gap indicates that it is lower.

A prevalence ratio above one indicates a higher estimate in the subgroup than in the reference group. A ratio below one indicates a lower estimate.

These measures describe marginal disparities. They do not represent age-adjusted racial/ethnic disparities or race-adjusted age disparities.

12.2 12.2. Survey design and uncertainty

The dataset contains published subgroup estimates rather than respondent-level observations and survey design information.

The analysis cannot reproduce the original survey weighting, stratification, clustering, or replicate-weight procedures.

Furthermore, the covariance between subgroup estimates is unavailable. Confidence intervals for individual estimates cannot be combined to produce exact confidence intervals for their differences without additional information.

Consequently, model-based uncertainty from the gap trajectories must be interpreted cautiously.

12.3 12.3. Temporal comparability

The HPS underwent changes in survey phases, frequency, methodology, and potentially measurement procedures.

The presence of a phase label does not by itself establish that a discontinuity occurred.

The official documentation must be reviewed before interpreting abrupt changes as substantive changes in population mental health.

12.4 12.4. Aggregate data limitations

The analysis uses aggregate subgroup estimates. It cannot:

- Track changes within individual respondents.
- Estimate individual-level risk factors.
- Adjust age disparities for race/ethnicity or vice versa.
- Establish causal effects of policy changes, vaccination, or pandemic events.
- Distinguish changes in symptom prevalence from all possible changes in survey response patterns.

12.5 12.5. Symptom indicators

The indicators measure symptoms consistent with anxiety or depressive disorders. They are not equivalent to clinical diagnoses.

12.6 12.6. General interpretation

The findings should be interpreted as changes in published population-level estimates and demographic gaps over time.

Any claims about narrowing or widening disparities should be supported by the estimated trajectories, model diagnostics, uncertainty, and sensitivity analyses.

13 13. Reproducibility and Export

13.1 13.1. Export model results

13.2 13.2. Record software versions

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19043)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: Asia/Tehran
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] broom_1.0.10    mgcv_1.9-3      nlme_3.1-168    scales_1.4.0
##  [5] knitr_1.50      janitor_2.2.1   lubridate_1.9.5 forcats_1.0.1
##  [9] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.0     readr_2.1.6
## [13] tidyr_1.3.2     tibble_3.3.0    ggplot2_4.0.1   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     stringi_1.8.7      lattice_0.22-7     hms_1.1.4
##  [5] digest_0.6.38      magrittr_2.0.4     evaluate_1.0.5     grid_4.5.2
##  [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-4
## [13] backports_1.5.0    cli_3.6.5          crayon_1.5.3       rlang_1.2.0
## [17] bit64_4.6.0-1      splines_4.5.2      withr_3.0.2        yaml_2.3.10
## [21] parallel_4.5.2     tools_4.5.2        tzdb_0.5.0         vctrs_0.7.3
## [25] R6_2.6.1           lifecycle_1.0.5    snakecase_0.11.1   bit_4.6.0
## [29] vroom_1.6.6        pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
## [33] glue_1.8.0         xfun_0.59          tidyselect_1.2.1   rstudioapi_0.17.1
## [37] farver_2.1.2       htmltools_0.5.8.1  labeling_0.4.3     rmarkdown_2.32
## [41] compiler_4.5.2     S7_0.2.1
```

14 14. References

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